U4: The Forced Structure of the Non-Relativistic Hamiltonian

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Abstract

We present the structural derivation of the non-relativistic single-particle Hamiltonian $\hat{H} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$ from the Event-Density (ED) primitive stack and the now-FORCED upstream items U1, U2, U5, and Theorem 10. Operating under Framing 1 (U4 conditional on the existence of a Hermitian Hamiltonian generator, which is sibling CANDIDATE U3's content), we decompose U4 into five sub-features — channel–momentum identification (F1), plane-wave eigenfunctions (F2), quadratic-in-$|\hat{p}|^2$ kinetic structure (F3), specific coefficient $1/(2m)$ (F4), and kinetic + potential decomposition (F5) — and derive each from primitive-level inputs plus standard mathematical infrastructure. F1 follows from Stone's theorem applied to translation symmetry on the U2 Hilbert space, transferring directly from the U5 derivation; F2 is mathematical; F5 follows from translation invariance + locality. F3 is established via four constraints (translation, rotation, analyticity, non-relativistic-limit), with all candidate alternatives — linear-in-$|\hat{p}|$, anisotropic, higher-even-power, full relativistic dispersion — explicitly dismissed within the non-relativistic scope. F4 is the methodologically distinctive content: **the factor of 2 in $1/(2m)$ is forced by integration of the Galilean commutator condition $[\hat{H}, \hat{K}_i] = -i\hbar\hat{p}_i$ against the boost generator $\hat{K} = m\hat{x} - t\hat{p}$, with the factor of 2 emerging as the integration Jacobian rather than as a convention borrowed from classical mechanics.** All seven falsifiers are dispatched within the stated scope conditions. The U4 verdict is **forced with form-forced-value-inherited framing**: the kinetic-energy form is uniquely derived from primitives + symmetry + scope; the mass parameter m is inherited per Arc M's "form FORCED, values INHERITED" methodology, and $\hbar$ is inherited via the dimensional-atlas Madelung anchoring. The arc introduces no new structural CANDIDATES; the conditionalities are all inherited. The active upstream-CANDIDATE inventory of the QM-emergence Phase-1 program reduces from {U3, U4} + continuum gauge to **{U3} + continuum gauge** — one upstream item remains to close the entire Phase-1 structural-foundations program. With U4 closed, the Phase-1 theorem inventory stands at 15 forced structural theorems; Schrödinger emergence is now FORCED conditional on U3 alone.

1 Introduction

The Schrödinger equation is one of the most directly tested structures in physics. Its non-relativistic single-particle form, $i\hbar\partial_t\psi = \hat{H}\psi$ with $\hat{H} = -\hbar^2\nabla^2/(2m) + V(\hat{x})$, has been used in calculations of atomic spectra, molecular bonding, semiconductor physics, neutron interferometry, and countless other empirical settings for nearly a century. The Hamiltonian's specific form — the squared-momentum kinetic energy with the specific factor $1/(2m)$, plus the additive position-dependent potential — is the structural anchor that connects the abstract evolution equation to physical observables.

In standard quantum mechanics, this Hamiltonian form is *imported* rather than derived. The kinetic-energy expression $T = p^2/(2m)$ is taken from classical mechanics, where it is the kinetic energy of a non-relativistic point particle; the operator-replacement $p \to \hat{p}$ promotes it to operator status. The factor of 2 in $1/(2m)$ is borrowed from the classical formula without further justification.

The Event-Density (ED) program asks whether this structural form can be derived rather than postulated. The QM-emergence Phase-1 framework [1] presents non-relativistic single-particle quantum mechanics as a structural consequence of a primitive ontology. Five arcs of the structural-foundations cycle (born_gleason [2], U2-Discrete [3], U2-Continuum [3], U5 [4], U1 [5]) have promoted the participation-measure carrier, its inner-product structure, the Born rule, the adjacency-band Fourier-conjugate partition, and the participation-measure construction itself from CANDIDATE to FORCED status. With those five arcs closed, two upstream commitments remained: U3 (the participation-measure evolution equation, supplying existence + linear-first-order structure of the time-evolution generator) and U4 (the specific Hamiltonian form, supplying the dynamical content of the kinetic energy). This paper presents the U4 arc and its verdict.

The U4 arc is the sixth and most delicate of the structural-foundations cycle. It is the first arc whose verdict is naturally framed as conditional on a *sibling* CANDIDATE (U3) — methodologically novel for the cycle — and it is the first arc whose load-bearing content concerns *dynamical structure* (the time-evolution generator's form) rather than kinematic structure. The arc's substantive structural finding is the derivation of the factor of 2 in $1/(2m)$ as an *integration Jacobian* of the Galilean Lie algebra commutator condition, rather than as a convention borrowed from classical mechanics. The arc closes FORCED at the form level, with the mass parameter m and the constant $\hbar$ inherited per established ED-program methodology.

The paper is organized as follows. Section 2 states the U4 question precisely. Section 3 inventories the structural inputs available under Framing 1. Section 4 decomposes U4 into five sub-features. Section 5 introduces the seven falsifiers. Sections 6–10 derive the five sub-features in turn, with Sections 9 and 10 carrying the load-bearing content. Section 11 consolidates falsifier dispatches. Section 12 documents the conditionality ledger. Section 13 states the final U4 theorem. Section 14 documents the downstream cascade. Section 15 discusses the result's significance.

2 Statement of the U4 Question

2.1 The target form

The QM-emergence Phase-1 framework's Step 2 (Schrödinger emergence) targets the standard non-relativistic single-particle Hamiltonian:

$$\hat{H} = \frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x}) \qquad \text{(U4)} \tag{1}$$

with the kinetic operator $T(\hat{p}) = \hbar^2|\hat{p}|^2/(2m)$ depending only on momentum and the potential operator $V(\hat{x})$ depending only on position. The participation-measure framework's evolution equation (U3) is

$$i\hbar\partial_t P_K = \hat{H}P_K + \sum_{K'} V_{KK'}P_{K'}.$$

U4 specifies the *form* of $\hat{H}$ entering this equation; U3 supplies the *existence + linear-first-order structure* of the equation itself.

2.2 The five sub-features

- **F1 (Channel–momentum identification).** In the thin/continuum limit, the channel index K admits a continuous momentum-space coordinate k via Stone's theorem on the participation graph's translation group.
- **F2 (Plane-wave eigenfunctions).** The eigenfunctions of the translation generator $\hat{p}$ in position representation are plane waves $\langle x|k\rangle = (2\pi\hbar)^{-d/2}e^{ikx/\hbar}$.

- **F3 (Quadratic-in-$|\hat{p}|^2$ kinetic structure).** $T(\hat{p}) \propto |\hat{p}|^2$, with linear-in-$|\hat{p}|$, higher-even-power, anisotropic, and full relativistic alternatives all excluded in the non-relativistic regime.
- **F4 (Specific coefficient $1/(2m)$).** $T(\hat{p}) = \hbar^2|\hat{p}|^2/(2m)$ — the factor of 2 in the denominator is structural, not conventional.
- **F5 (Kinetic + potential decomposition).** $\hat{H} = T(\hat{p}) + V(\hat{x})$ with no cross terms, in the gauge-coupling-free scope.

2.3 The seven falsifiers

- **Fal-1.** No momentum-basis identification.
- **Fal-2.** Linear-in-$|\hat{p}|$ kinetic term (e.g., $T = c|\hat{p}|$, photon-like dispersion).
- **Fal-3.** Higher-even-power terms (e.g., $T \supset c|\hat{p}|^4$ in the non-relativistic regime).
- **Fal-4.** Coefficient $\neq 1/(2m)$ — wrong factor.
- **Fal-5.** Anisotropic kinetic energy (direction-dependent).
- **Fal-6.** Position-momentum cross terms in $\hat{H}$.
- **Fal-7.** Full relativistic dispersion in non-relativistic regime.

2.4 Framing 1: U4 conditional on sibling candidate U3

The U4 arc adopts **Framing 1**: U4 is derived *given* that a Hermitian Hamiltonian generator $\hat{H}$ exists on the participation-measure Hilbert space. The existence question is U3's content; U4 specifies the form. This framing is methodologically novel within the structural-foundations cycle.

Under Framing 1, U3 cannot be invoked as a *derivation input* for U4 (that would be circular), but the existence of a Hermitian generator can be assumed as the starting point. The verdict is then "U4 FORCED conditional on U3"; promoting U3 in a subsequent arc would make U4 fully unconditional.

3 Structural Inputs

3.1 Primitives

- **Primitive 03 (Participation):** graph homogeneity, supplying translation symmetry kinematically.
- **Primitive 06 (ED Gradient):** $\nabla\rho$ supplies a spatial direction with no preferred origin.
- **Primitive 09 (Tension Polarity):** $U(1)$-valued polarity phase entering the participation-measure construction via U1.
- **Primitive 13 (Relational Timing):** time-axis structure, used in F4 to combine with spatial translation symmetry into the full Galilean group.

3.2 Inherited forced upstream items

The U4 arc has access to all five FORCED items from prior arcs of the structural-foundations cycle: U1 [5], U2-Discrete and U2-Continuum [3], U5 [4], Theorem 10 [2]. Particularly, the Stone's-theorem derivation in U5 transfers directly to F1 of the present arc.

3.3 Mathematical infrastructure

- Stone's theorem on one-parameter unitary groups (Stone 1932) [6].
- Standard functional calculus on self-adjoint operators.
- Galilean Lie algebra and central extension (Bargmann 1954) [7].
- Standard Fourier transform on $L^2(\mathbb{R}^d)$.

3.4 Scope conditions

Two scope conditions inherited from Phase-1 Step 2's framing:

- **Non-relativistic single-particle scope.** $|\hat{p}|^2/(mc)^2 \ll 1$.
- **Gauge-coupling-free scope.** Magnetic vector potentials would couple position and momentum via $T(\hat{p} - eA(\hat{x})/c)$; F5 holds in the absence of such gauge couplings.

3.5 Inherited values

- **Mass m:** per Arc M's "form FORCED, values INHERITED" methodology [8].
- **$\hbar$:** inherited via the dimensional-atlas Madelung anchoring.

3.6 Forbidden inputs (under Framing 1)

The following inputs are explicitly forbidden as derivation inputs to U4: U3 (sibling CANDIDATE; may be invoked only as conditional); the classical Hamiltonian as a postulate; the Schrödinger equation in textbook form; the relativistic dispersion as a derivation input.

4 Sub-Feature Decomposition

Sub-feature	Status	Structural inputs	Falsifiers
F1	FORCED	Primitives 03 + 06 + U2 + Stone's theorem; U3-independent	Fal-1
F2	FORCED (automatic given F1)	F1 + standard math	—
F3	FORCED	C1 translation + C2 rotation + C3 analyticity + C4 non-rel limit	Fal-2, Fal-3, Fal-5, Fal-7
F4	FORCED form (factor of 2 derived); m, $\hbar$ INHERITED	Primitives 03 + 13 + non-rel scope + U2 unitarity + Galilean Lie algebra integration	Fal-4
F5	FORCED (gauge-coupling-free scope)	Translation invariance + locality	Fal-6

5 Falsifier Inventory

Each falsifier targets a specific physical alternative:

- **Fal-1:** Channel index has no momentum-space interpretation. Tested by Stone's theorem on the U2 Hilbert space.
- **Fal-2:** Massless photon-like dispersion in a regime supposed to describe massive particles. Tested by C2 + C3 + C4.

- **Fal-3:** Relativistic corrections in non-relativistic regime. Tested by C4 scope condition.
- **Fal-4:** Wrong proportionality between kinetic energy and momentum-squared. Tested by Galilean Lie algebra integration.
- **Fal-5:** Anisotropic kinetic energy. Tested by C2 rotation invariance.
- **Fal-6:** Position-momentum cross terms. Tested by translation invariance + locality.
- **Fal-7:** Full relativistic dispersion. Tested by C4 non-relativistic-limit.

6 Derivation of F1 (Momentum Identification)

6.1 The claim

In the thin/continuum limit, the channel index K admits a continuous momentum-space coordinate k via Stone's theorem on the participation graph's translation group, with the basis-change between channel-representation and momentum-representation the standard Fourier transform.

6.2 Translation symmetry as a kinematic property

Define spatial translation T_a acting on the participation-measure space by $(T_a P)_K(u) := P_K(u-a)$. Primitive 03's homogeneity guarantees that the participation relation does not privilege any vertex. The set $\{T_a : a \in \mathbb{R}^d\}$ forms a d-parameter abelian Lie group.

6.3 Translation acts unitarily on the U2 Hilbert space

For any $P, Q \in \mathcal{H}$ and any translation a:

$$\langle T_a P \mid T_a Q \rangle = \sum_K \int du\, P_K^*(u-a)\, Q_K(u-a) = \sum_K \int du'\, P_K^*(u')\, Q_K(u') = \langle P \mid Q \rangle$$

under change of variables $u' = u - a$. The translation group preserves the U2 inner product. Continuity follows from continuity of the participation-measure field. Therefore $\{T_a\}$ is a strongly continuous one-parameter unitary group on $\mathcal{H}$.

6.4 Stone's theorem identifies the translation generator

Theorem 6.1 (Stone 1932 [6])**.** *Every strongly continuous one-parameter unitary group on a Hilbert space has a unique self-adjoint generator.*

Applied to the spatial translation group:

$$T_a = \exp\left(\frac{i\hat{p} \cdot a}{\hbar}\right). \tag{2}$$

The eigenstates $|k\rangle$ of $\hat{p}$ form a complete orthonormal basis on $\mathcal{H}$. **In the thin/continuum limit, the channel index K is identified with the momentum eigenvalue k:**

$$K \leftrightarrow k. \tag{3}$$

6.5 Explicit U3-independence

The derivation invokes Primitives 03 + 06, U2, and Stone's theorem. **No time-evolution operator, no Hamiltonian, no Schrödinger equation appears in the argument.** Stone's theorem identifies $\hat{p}$ kinematically on the U2 Hilbert space.

Theorem 6.2 (F1)**.** *Let $\mathcal{H}$ be the U2 Hilbert space of participation measures. Let $\{T_a : a \in \mathbb{R}^d\}$ be the spatial translation group acting on $\mathcal{H}$. Then $\{T_a\}$ is a strongly continuous one-parameter unitary group, and there exists a unique self-adjoint operator $\hat{p}$ such that $T_a = \exp(i\hat{p} \cdot a/\hbar)$. The eigenstates of $\hat{p}$ form a complete orthonormal basis on $\mathcal{H}$, and in the thin/continuum limit the channel index K is identified with the eigenvalue k of $\hat{p}$.*

Falsifier Fal-1 dispatched.

7 Derivation of F2 (Plane-Wave Eigenfunctions)

7.1 Position-representation form of $\hat{p}$

Given F1's identification of $\hat{p}$, the position-representation form is determined by:

$$T_a\psi(x) = \psi(x-a) = \exp(-a \cdot \nabla)\,\psi(x).$$

Comparing with (2):

$$\hat{p} = -i\hbar\nabla. \tag{4}$$

7.2 Eigenfunctions

The eigenvalue equation $\hat{p}\,|k\rangle = k\,|k\rangle$ becomes, in position representation, $-i\hbar\nabla\langle x|k\rangle = k\langle x|k\rangle$, with solution

$$\langle x|k\rangle = (2\pi\hbar)^{-d/2} \cdot e^{ik\cdot x/\hbar}. \tag{5}$$

The normalization $(2\pi\hbar)^{-d/2}$ is fixed by the orthonormality $\langle k|k'\rangle = \delta(k-k')$.

Theorem 7.1 (F2)**.** *In the position representation of the U2 Hilbert space, the translation generator $\hat{p}$ takes the form $\hat{p} = -i\hbar\nabla$. The eigenfunctions of $\hat{p}$ are plane waves $\langle x|k\rangle = (2\pi\hbar)^{-d/2} \cdot e^{ik\cdot x/\hbar}$.*

F2 introduces no new structural commitment beyond F1.

8 Derivation of F5 (Kinetic + Potential Decomposition)

8.1 Translation invariance forces the free part to depend only on $\hat{p}$

In the absence of an external potential, $\hat{H}_{\text{free}}$ commutes with translations: the participation graph itself is translation-invariant (Primitive 03), and without external coupling, $\hat{H}_{\text{free}}$ inherits this invariance. Any operator commuting with $\hat{p}$ depends on $\hat{p}$ alone via standard functional calculus. Therefore

$$\hat{H}_{\text{free}} = T(\hat{p}). \tag{6}$$

8.2 Locality forces external potentials to depend only on $\hat{x}$

Locality is a primitive-level structural property of the participation graph: edges connect adjacent vertices, and external influences couple at specific locations. The local action of a potential takes the form $(V\psi)(x) = V(x)\psi(x)$ — multiplication by a position-dependent function, depending only on $\hat{x}$.

A potential depending on $\hat{p}$ would be non-local in position representation and is structurally excluded from the standard non-relativistic-single-particle scope.

8.3 Cross terms forbidden

Any cross term $\hat{p}\hat{x}$ or $\hat{x}\hat{p}$ violates either translation invariance (free part) or locality (potential part).

8.4 Scope caveat

In the presence of magnetic vector potentials $A(\hat{x})$, the kinetic operator becomes $T(\hat{p} - eA(\hat{x})/c)$. The scope condition for U4's F5 is non-relativistic single-particle Schrödinger emergence *without* gauge couplings.

Theorem 8.1 (F5)**.** *Within the gauge-coupling-free non-relativistic single-particle scope of Phase-1 Step 2, the Hamiltonian on the U2 Hilbert space admits the unique additive decomposition $\hat{H} = T(\hat{p}) + V(\hat{x})$, with the kinetic part depending only on $\hat{p}$ and the potential part depending only on $\hat{x}$. Cross terms are forbidden.*

Falsifier Fal-6 dispatched.

9 Derivation of F3 (Quadratic Kinetic Structure)

9.1 The four constraints

- **C1 Translation invariance.** $T = T(\hat{p})$ alone (already established).
- **C2 Rotation invariance.** The participation graph supports rotations as a kinematic symmetry.
- **C3 Analyticity.** $T(\hat{p}) = f(|\hat{p}|^2)$ is analytic in $|\hat{p}|^2$ at $\hat{p} = 0$.
- **C4 Non-relativistic-limit scope.** $|\hat{p}|^2/(mc)^2 \ll 1$.

9.2 C1 + C2 reduce to $f(|\hat{p}|^2)$

By C1, $T = T(\hat{p})$. By C2, T is invariant under rotations of $\hat{p}$. The unique rotational invariant of $\hat{p}$ (up to powers) is $|\hat{p}|^2$:

$$T(\hat{p}) = f(|\hat{p}|^2) \tag{7}$$

for some function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$. **Anisotropic alternatives dismissed — Fal-5 dispatched.**

9.3 C3 forces Taylor expansion with no half-integer powers

By C3:

$$f(|\hat{p}|^2) = a_0 + a_1|\hat{p}|^2 + a_2(|\hat{p}|^2)^2 + a_3(|\hat{p}|^2)^3 + \dots \tag{8}$$

with $a_0 = 0$ by convention.

A candidate term $a_{1/2}|\hat{p}| = a_{1/2}\sqrt{|\hat{p}|^2}$ is *not analytic* in $|\hat{p}|^2$ at the origin (branch point) and is excluded by C3. The vector linear $\alpha \cdot \hat{p}$ is excluded by C2. **Falsifier Fal-2 dispatched.**

9.4 C4 suppresses higher-even-power terms

In the non-relativistic regime, with the natural dimensional scaling $a_n \sim a_1/(mc)^{2(n-1)}$:

$$\frac{a_n(|\hat{p}|^2)^n}{a_1|\hat{p}|^2} \sim \left(\frac{|\hat{p}|^2}{(mc)^2}\right)^{n-1} = \left(\frac{v}{c}\right)^{2(n-1)}.$$

Higher-order terms are suppressed by powers of $(v/c)^2$ and vanish in the strict $c \to \infty$ limit. Only $a_1|\hat{p}|^2$ survives. **Falsifier Fal-3 dispatched.**

9.5 Relativistic dispersion in non-relativistic regime

The full relativistic dispersion $T_{\mathrm{rel}}(\hat{p}) = \sqrt{|\hat{p}|^2c^2 + m^2c^4} - mc^2$ Taylor-expands as

$$T_{\mathrm{rel}}(\hat{p}) = \frac{|\hat{p}|^2}{2m} - \frac{|\hat{p}|^4}{8m^3c^2} + O\left(\frac{|\hat{p}|^6}{m^5c^4}\right).$$

In the strict non-relativistic limit, only the leading $|\hat{p}|^2/(2m)$ term survives. **Falsifier Fal-7 dispatched.**

Theorem 9.1 (F3)**.** *In the non-relativistic single-particle gauge-coupling-free scope, the kinetic-energy operator $T(\hat{p}) = f(|\hat{p}|^2)$ on the U2 Hilbert space takes the unique form*

$$T(\hat{p}) = a_1|\hat{p}|^2$$

with $a_1 > 0$ a real constant determined by F4. All candidate alternatives — anisotropic, linear-in-$|\hat{p}|$, higher-even-power, full relativistic — are excluded by C1 + C2 + C3 + C4.

Falsifiers Fal-2, Fal-3, Fal-5, Fal-7 dispatched.

10 Derivation of F4 (Coefficient $1/(2m)$)

10.1 The factor-of-2 question

Dimensional analysis fixes $a_1 = c_1\hbar^2/m$ with c_1 dimensionless. The standard non-relativistic value is $c_1 = 1/2$, but dimensional analysis alone does not fix it. **The factor of $1/2$ is the load-bearing question** — to be derived from primitive-level inputs, not borrowed by convention from classical mechanics.

10.2 Galilean Lie algebra

The Galilean Lie algebra is generated by spatial translations $\hat{P}_i$, time translation $\hat{H}$, rotations $\hat{J}_i$, and Galilean boosts $\hat{K}_i$. The structurally substantive commutators are:

$$[\hat{K}_i, \hat{P}_j] = i\hbar m\,\delta_{ij}, \tag{9}$$

$$[\hat{H}, \hat{K}_i] = -i\hbar\hat{p}_i. \tag{10}$$

The mass m enters as the central-extension structure constant. The boost generator takes the form

$$\hat{K}_i = m\hat{x}_i - t\hat{p}_i. \tag{11}$$

10.3 Galilean uniqueness within non-relativistic scope

Within the non-relativistic scope (C4 + absolute time), the Galilean group is the unique consistent boost-translation algebra. Lorentzian boosts mix space and time (relativistic regime); Galilean boosts preserve absolute time (non-relativistic regime). The non-relativistic-limit scope condition selects Galilean over Lorentzian.

10.4 Galilean transformations act unitarily on the U2 Hilbert space

- Translations $\hat{p}$: unitary by F1 (Stone's theorem).
- Time translations $\hat{H}$: assumed self-adjoint under Framing 1.
- Rotations $\hat{J}$: unitary by C2.
- Boosts $\hat{K} = m\hat{x} - t\hat{p}$: self-adjoint as real-coefficient linear combination of self-adjoint operators.

All Galilean transformations are unitary on the U2 Hilbert space.

10.5 Momentum transformation under boosts

Under an infinitesimal Galilean boost with velocity v:

$$\hat{p}'_j = \hat{p}_j + \frac{iv_i}{\hbar}[\hat{K}_i, \hat{p}_j] + O(v^2) = \hat{p}_j - mv_j + O(v^2).$$

For a finite boost:

$$\hat{p} \to \hat{p} - mv. \tag{12}$$

10.6 The integration that fixes the factor of 2

The Hamiltonian $\hat{H} = T(\hat{p})$ must satisfy (10). Substituting (11) and using $[\hat{H}, \hat{p}_i] = 0$:

$$[\hat{H}, \hat{K}_i] = m[T(\hat{p}), \hat{x}_i] = -i\hbar m \frac{\partial T(\hat{p})}{\partial \hat{p}_i}.$$

Setting equal to $-i\hbar\hat{p}_i$:

$$\frac{\partial T(\hat{p})}{\partial \hat{p}_i} = \frac{\hat{p}_i}{m}. \tag{13}$$

Using the F3 result $T = f(|\hat{p}|^2)$, so $\partial T/\partial \hat{p}_i = 2\hat{p}_i f'(|\hat{p}|^2)$:

$$2\hat{p}_i f'(|\hat{p}|^2) = \frac{\hat{p}_i}{m} \quad \Longrightarrow \quad f'(|\hat{p}|^2) = \frac{1}{2m}.$$

Integrating with $f(0) = 0$:

$$f(|\hat{p}|^2) = \frac{|\hat{p}|^2}{2m}. \tag{14}$$

The factor of 2 emerges from the integration. The coefficient $1/(2m)$ in $T = |\hat{p}|^2/(2m)$ comes from integrating $f'(|\hat{p}|^2) = 1/(2m)$ — the factor of 2 is the chain-rule *Jacobian* relating $\partial T/\partial \hat{p}_i = 2\hat{p}_i f'(|\hat{p}|^2)$ to the Galilean commutator condition. **It is not a convention; it is the result of integrating the differential equation that Galilean invariance demands.**

10.7 Form-FORCED, value-INHERITED

Combining F3 and F4, restoring $\hbar$:

$$T(\hat{p}) = \frac{\hbar^2|\hat{p}|^2}{2m} \tag{15}$$

with m inherited per Arc M and $\hbar$ inherited per the dimensional-atlas Madelung anchoring. **Falsifier Fal-4 dispatched.**

Theorem 10.1 (F4)**.** *Under Framing 1 and within the non-relativistic single-particle scope, the kinetic-energy operator $T(\hat{p}) = a_1|\hat{p}|^2$ established by F3 has*

$$a_1 = \frac{\hbar^2}{2m}$$

uniquely. The factor of 2 emerges from integrating the Galilean commutator condition $[\hat{H}, \hat{K}_i] = -i\hbar\hat{p}_i$ against $\hat{K} = m\hat{x} - t\hat{p}$, with the factor of 2 the chain-rule Jacobian. The mass m is inherited per Arc M; $\hbar$ is inherited via the dimensional-atlas Madelung anchoring.

11 Falsifier Resolution Table

Falsifier	Target	Dispatch argument	Memo
Fal-1	F1	Stone's theorem on U2 Hilbert space's translation group produces unique self-adjoint $\hat{p}$	Memo 02 §1
Fal-2	F3	Vector $\alpha \cdot \hat{p}$ excluded by C2; scalar $\lvert\hat{p}\rvert c$ excluded by C3 (branch-point) + C4	Memo 03 §2.4
Fal-3	F3	C4 non-rel limit suppresses all $a_n \lvert\hat{p}\rvert^{2n}$ for $n \geq 2$ by $(v/c)^{2(n-1)}$	Memo 03 §2.5
Fal-4	F4	Galilean commutator integration: $f'(\lvert\hat{p}\rvert^2) = 1/(2m)$; factor of 2 is integration Jacobian	Memo 03 §3.7
Fal-5	F3	C2 forces $T(\hat{p}) = f(\lvert\hat{p}\rvert^2)$	Memo 03 §2.3
Fal-6	F5	Translation invariance + locality forbid cross terms (gauge-coupling-free scope)	Memo 02 §3
Fal-7	F3	C4 scope: relativistic dispersion reduces to U4 form in $c \to \infty$ limit	Memo 03 §2.6

All seven falsifiers dispatched within the stated scope conditions.

12 Conditionality Ledger

The U4 verdict is FORCED, conditional on:

1. **U3 (sibling candidate; Framing 1).** U3 supplies the existence of a Hermitian Hamiltonian generator. U4 specifies the form. Promoting U3 makes U4 fully unconditional.

2. **Non-relativistic single-particle scope.** Inherited from Phase-1 Step 2; conditions F3 and F4.

3. **Gauge-coupling-free scope.** F5's clean decomposition; gauge field theory is downstream content.

4. **Inherited values.** Mass m per Arc M; $\hbar$ per dimensional-atlas Madelung.

U4 introduces no new structural candidates. All four conditionalities are inherited from prior structural commitments.

13 Final U4 Theorem

Theorem 13.1 (U4; Specific Hamiltonian Form)**.** *Let the ED primitive stack supply: bandwidth (Primitive 04), spatial structure (Primitive 06), participation relations (Primitive 03), channel index (Primitive 07), tension polarity (Primitive 09), and relational timing (Primitive 13), together with the now-FORCED upstream items U1, U2-Discrete, U2-Continuum, U5, and Theorem 10. Within the non-relativistic single-particle scope of Phase-1 Step 2 (gauge-coupling-free), and conditional on the existence of a Hermitian Hamiltonian generator (sibling CANDIDATE U3, Framing 1), the Hamiltonian operator on the participation-measure Hilbert space takes the unique form*

$$\hat{H} = \frac{\hbar^2 |\hat{p}|^2}{2m} + V(\hat{x})$$

with the following structural properties:

1. ***Channel–momentum identification.*** *In the thin/continuum limit, $K \leftrightarrow k$ via Stone's theorem.*

2. ***Kinetic + potential decomposition.*** *$\hat{H} = T(\hat{p}) + V(\hat{x})$; cross terms forbidden.*

3. ***Quadratic-in-$|\hat{p}|^2$ kinetic structure.*** *$T(\hat{p}) = a_1|\hat{p}|^2$ uniquely.*

4. ***Specific coefficient $\hbar^2/(2m)$.*** *$a_1 = \hbar^2/(2m)$, with the factor of 2 forced by Galilean commutator integration. m inherited per Arc M; $\hbar$ inherited per dimensional-atlas Madelung.*

5. ***Form $V(\hat{x})$.*** *Scalar function of position; specific V inherited from external context.*

Status: FORCED with form-FORCED-value-INHERITED framing.

14 Downstream Consequences

14.1 Updated structural-foundations ledger

#	Theorem	Status
10	Born rule (Gleason–Busch)	FORCED
11	U2-Discrete	FORCED
12	U2-Continuum	FORCED (with conformal gauge)
13	U5	FORCED
14	U1	FORCED
15	U4 (this paper)	FORCED (form; values inherited)

14.2 Active CANDIDATE inventory

Pre-U4	Post-U4
{U3, U4} + continuum gauge	{U3} + continuum gauge

One upstream CANDIDATE remains. Promoting U3 closes the QM-emergence Phase-1 program except for the description-level continuum gauge convention.

14.3 Schrödinger emergence

The Phase-1 Step 2 chain reads:

```
Primitives 03, 04, 06, 07, 09, 13
              |
              v
        U1 (FORCED) -> P_K = sqrt(b) * exp(i pi)
              |
              v
        U2 (FORCED) -> <P|Q> inner product
              |
              v
        U5 (FORCED) -> adjacency partition b = b_x + b_p
              |
              v
        U4 (FORCED, this paper) -> H = h^2 |p|^2/(2m) + V(x)
              |
              v
        U3 (CANDIDATE) -> i h dt P = H P
              |
              v
      Schrodinger emergence (Step 2)
```

Schrödinger evolution is forced conditional on U3 alone.

14.4 Preparation for the U3 arc

The U3 arc would address: existence of self-adjoint $\hat{H}$ via Stone's theorem on time-translation symmetry (Primitive 13); linearity; first-order-in-time character. The U3 arc inherits all six FORCED upstream items. Anticipated arc structure: 4 memos parallel to U4.

15 Discussion and Outlook

15.1 Position in the Phase-1 program

U4 is the sixth and most delicate arc of the structural-foundations cycle. The cycle (born_gleason → U2-Discrete → U2-Continuum → U5 → U1 → U4) has demonstrated the structural-derivation methodology against six substantively different structural questions, including the dynamical-content question handled here.

15.2 Why the factor-of-2 result matters

The factor of 2 in $1/(2m)$ has appeared in classical mechanics for centuries. In standard quantum mechanics, it is borrowed from the classical formula via $p \to \hat{p}$. The connection is left as a postulate.

The U4 result establishes that this connection is structurally derivable. The factor of 2 is forced by integration of the Galilean Lie algebra commutator condition — the integration Jacobian of the differential equation $\partial T/\partial \hat{p}_i = \hat{p}_i/m$ produced by the Galilean commutator condition. The factor is not a feature of classical mechanics that quantum mechanics happens to inherit; it is a feature of *Galilean kinematics* that both classical and quantum mechanics inherit.

15.3 How U4 + U5 + U2 form the dynamical backbone

The structural-foundations cycle decomposes into three layers:

- **Layer 1 (carrier):** U1.
- **Layer 2 (Hilbert-space arena):** U2-Discrete + U2-Continuum + Theorem 10 + U5.
- **Layer 3 (dynamical):** U3 + U4.

U2 + U5 + U4 supply the kinematic + dynamical structural backbone of the Schrödinger equation. With U3 closed in the next arc, the full dynamical equation would be structurally complete.

15.4 Next steps

1. Open the U3 arc.
2. Revise the QM-emergence Phase-1 synthesis paper.
3. Draft the U5 publication paper (the missing fourth in the structural-foundations publication series).

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Manuscript prepared April 2026. Source materials: `arcs/U4/` four-memo arc and closure memo, integrated into a single publishable document. The U4 arc is the sixth and most delicate of the structural-foundations cycle.
Draft complete. Pausing for revision.